APTITUDE

Aptitude Questions & Solved Problems

Comprehensive Question Bank: Quantitative Aptitude

1. Number System & Simplification

- Rule of BODMAS: Brackets -> Order -> Division -> Multiplication -> Addition -> Subtraction.
- Problem 1: Find the value of 1800 ÷ [10{(12-6) + (24-12)}].
  Solution: Inside innermost brackets: 12-6=6, 24-12=12. Sum=18. 10*18=180. 1800 ÷ 180 = 10.
- Problem 2: If a number is divided by 136, it leaves 36 as remainder. What is the remainder when divided by 17?
  Solution: 136 is divisible by 17 (17*8 = 136). Remainder = 36 mod 17 = 2.

2. Percentage & Profit/Loss

- Percentage Increase = (Increase / Original) * 100.
- Problem: In an exam 65% of students passed and 420 failed. What is the total strength?
  Solution: Failed percentage = 35%. 0.35 * Total = 420 => Total = 420 / 0.35 = 1,200 students.
- Problem: Selling an article at Rs. 990 yields 10% loss. What selling price gives 10% profit?
  Solution: Cost Price = 990 / 0.9 = Rs. 1,100. Target SP = 1,100 * 1.10 = Rs. 1,210.

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Time, Work, Pipes & Cisterns

3. Time & Work Core Equations

- Work done per day by person A taking 'a' days = 1/a.
- Combined work rate: If A takes 10 days and B takes 15 days, 1/Total = 1/10 + 1/15 = 5/30 = 1/6.
  Total combined time = 6 days.
- Efficiency principle: If Tanya is 25% more efficient than Sakshi (20 days):
  Efficiency ratio = 1.25 : 1 => Time taken by Tanya = 20 / 1.25 = 16 days.

4. Speed, Distance & Relative Motion

- Conversion factor: km/hr to m/s = Multiply by 5/18.
- Problem: A train 150m long moving at 60 km/h crosses a 300m platform.
  Total distance = 150 + 300 = 450m. Speed = 60 * (5/18) = 50/3 m/s.
  Time required = Distance / Speed = 450 / (50/3) = 27 seconds.